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Polyiamonds Attaining Extremal Topological Properties

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Abstract

We consider two optimization questions with respect to polyiamonds. What is the maximum number of holes that a polyiamond with n tiles can enclose, and what is the minimum number of tiles required to construct a polyiamond with h holes? These numbers will be given by the sequences $f_\triangle(n)$ and $g_\triangle(h)$, respectively. We are able to completely answer these questions and give the values of these sequences for all $n, h \geq 1$, in part by constructing a sequence of polyiamonds which has h holes and $g_\triangle(h)$ tiles for every $h \geq 0$. Such a polyiamond is called *crystallized*.

1 Introduction

A polyiamond is a planar shape formed by gluing together a finitely many congruent equilateral triangles along their edges. If two triangles of a polyiamond intersect, then their intersection is an entire edge, and the gluing requirement implies that a polyiamond must have connected interior. We only consider *free* polyiamonds; that is two polyiamonds are equivalent if they agree up to isometry. We refer to the triangles on a polyiamond as either triangles or tiles.

One longstanding open problem in enumerative combinatorics is determining the number of polyiamonds with exactly n tiles. We denote the set of all polyiamonds with n tiles by $\mathcal{T}_n$, the cardinality of $\mathcal{T}_n$ by $a(n)$, and the number of tiles in a polyiamond A by $|A|$. The values of the sequence $a(n)$ are

known up to $n \leq 75$ [4]. It is known that the sequence grows exponentialy fast and that its asymptotic growth constant is equal to $\lim_{n\to\infty} a(n+1)/a(n)$ (see [1] and [2] for the most recent improvements on the upper and lower bounds for this constant).

In the present paper, we study the extremal topological combinatorics problems of maximizing the number of holes for a polyiamond with n tiles, and conversely, of minimizing the number of tiles required to create a polyiamond with h holes.

To be precise about the topology, we consider the tiles of a polyiamond to be closed. Polyiamonds are finite unions of these closed tiles, so they are compact. The holes of a polyiamond are the bounded, connected components of the polyiamond's complement in the plane, and the number of holes is the number of those components minus one (the infinite component). The area of a hole is defined to be the number of tiles needed to fill it in.

Given a polyiamond A, we denote by $h(A)$ the number of holes in A. In Figure 1, we show polyiamonds with 1, 2, and 3 holes, respectively. These polyiamonds are optimal in the sense that it is impossible to create more holes with the same number of tiles, and it is impossible to create these numbers of holes with fewer tiles. These two topological and geometric properties are precisely defined in the next two definitions.

Definition 1.1. For $n \geq 1$, we define the sequence of the maximum number of holes that a polyiamond with n tiles can have as

$$f_{\triangle}(n) := \max_{A \in \mathcal{T}_n} h(A). \tag{1}$$

Definition 1.2. For $h \geq 1$, we define the sequence of the minimum number of tiles needed for constructing a polyiamond with h holes as

$$g_{\triangle}(h) := \min_{h(A)=h} |A|. \tag{2}$$

T_1 T_2 T_3

Figure 1: Polyiamonds with the minimum number of tiles for up to three holes.

As far as we know, this is the first time that the sequences $f_\triangle(n)$ and $g_\triangle(h)$ are being defined and studied.

1.1 Statement of Main Result

Here and throughout,

$$h_k = 3\binom{k}{2} = \frac{3}{2}k^2 - \frac{3}{2}k, \qquad n_k = \frac{9}{2}k^2 + \frac{3}{2}k - 2. \tag{3}$$

Theorem 1.1. *The first three values of $g_\triangle$ are $g_\triangle(1) = 9$, $g_\triangle(2) = 14$, and $g_\triangle(3) = 19$. Then for $h_k \leq h \leq h_{k+1}$ for a fixed $k \geq 2$,*

$$g_\triangle(h) = 3h + 3k + 1 + \left\lceil \frac{2h}{k} \right\rceil. \tag{4}$$

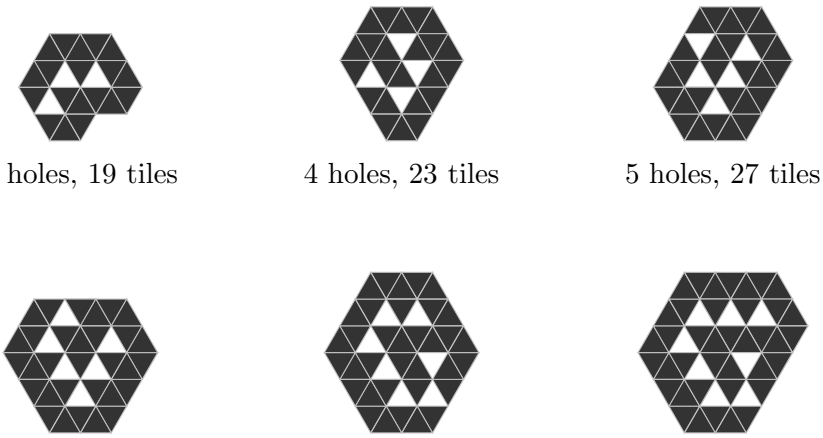
At the endpoints $g_\triangle(h_k) = n_k$, and for $h \geq 3$ the values of $g_\triangle(h)$ increase incrementally by either three or four, the latter only at h such that $\lceil 2h/k \rceil$ jumps by 1. It is clear from Definition 1.1 that $f_\triangle(n) = \max\{h : g_\triangle(h) \leq n\}$, thus Theorem 1.1 completely determines $f_\triangle(n)$. In particular, as a direct result of the property that $g_\triangle(h+1) - g_\triangle(h) \geq 1$, $f_\triangle$ is a step function which is constant on the intervals $g_\triangle(h) \leq n < g_\triangle(h+1)$.

Theorem 1.2. *For all $n \geq 1$, $f_\triangle(n) = h$ where $g_\triangle(h) \leq n < g_\triangle(h+1)$.*

Consequently, we also see that $f_\triangle(h+1) - f_\triangle(h) \leq 1$. While this follows incidentally here by indirect and primarily arithmetic methods, the fact that the increments in $f_\triangle$ are never more than one is not at all trivial. It may well be, *a priori*, that adding a single tile to an optimal configuration allows for rearrangements which produce several new holes at once. A purely geometric proof that this cannot happen in polyiamonds can be obtained via similar methods to those used in [7] with polyominoes, in which the tiles are unit squares instead of triangles.

Since we use the values of $g_\triangle$ to determine $f_\triangle$, we are most interested in polyiamonds which have h holes and $g_\triangle(h)$ tiles.

Definition 1.3. A polyiamond A with n tiles and h holes is *crystallized* if $g_\triangle(h(A)) = |A|$.



3 holes, 19 tiles 4 holes, 23 tiles 5 holes, 27 tiles

6 holes, 31 tiles 7 holes, 35 tiles 8 holes, 39 tiles

Figure 2: Crystallized polyiamonds for $3 \leq h \leq 8$.

A natural question to ask is if crystallized polyiamonds are unique. In almost all cases, the answer is no. However, their structure is rather concisely determined for all h, given by the characterization in Theorem 1.4.

Theorem 1.3. *For every positive integer $h \geq 3$ there exists at least two distinct crystallized polyiamonds with h holes.*

Theorem 1.4. *If A is a crystallized polyiamond, then the dual graph of A is a tree, the outer perimeter of A fills out a shape of minimum perimeter for $|A| + h(A)$ many tiles, and every hole in A has an area of 1.*

In contrast to this, in polyominoes few exact values are known for the sequences $f_\square(n)$ and $g_\square(h)$, and there are no analogous structural theorems in place. With exhaustive computational work in [3] in 2015, Tomás Oliveira e Silva enumerated all polyominoes with less than 28 tiles and determined $f_\square(n)$ for $n \leq 28$ and $g_\square(h)$ for $h \leq 8$.

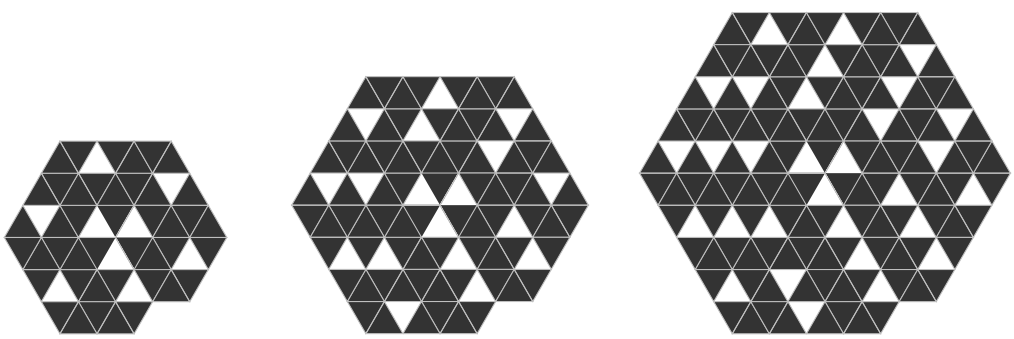
In 2019, Kahle and Roldán were the first to explicitly study these sequences for polyominoes in [7], establishing the first and second order asymptotics of $f_\square(n)$ and obtaining the values of $g_\square(h)$ for a sequence with exponential growth. In particular, they show that given $C_1 > \sqrt{5/2}$ and $C_2 < \sqrt{3/2}$, there exists an $n_0 = n_0(C_1, C_2)$ such that

$$\frac{1}{2}n - C_1\sqrt{n} \leq f(n) \leq \frac{1}{2}n - C_2\sqrt{n},$$

for $n > n_0$; and for

$$s_k = \frac{1}{3}\left(2^{2k} - 1\right), \qquad l_k = \frac{1}{3}\left(2^{2k+1} + 3 \cdot 2^{k+1} + 3\right),$$

it is the case that $g_{\square}(s_k) = l_k$ for all integers $k \geq 1$. They construct highly symmetric crystallized polyominoes attaining these values (Figures 1 and 5 in [7]), which also satisfy structural requirements analogous to Theorem 1.4. It is an open question as to whether these requirements hold for all crystallized polyominoes.



T_9 T_{18} T_{30}

Figure 3: Crystallized polyiamonds in $\{T_h\}$ with 9, 18, and 30 holes.

In proving Theorem 1.1 in Section 3, we iteratively construct a sequence of crystallized polyiamonds $\{T_h\}_{h\geq 1}$, with $h(T_h) = h$. The first three in this sequence are depicted in Figure 1, and for $h \in \{h_3, h_4, h_5\}$ in Figure 3. These polyiamonds have essentially a three-fold symmetry at rotations of $2\pi/3$, with zig-zagging arrays of tiles at angles $0, 2\pi/3$, and $4\pi/3$, and straight lines breaking off from these at angles of $\pi/3$, π, and $5\pi/3$. This is especially notable in Figure 4.

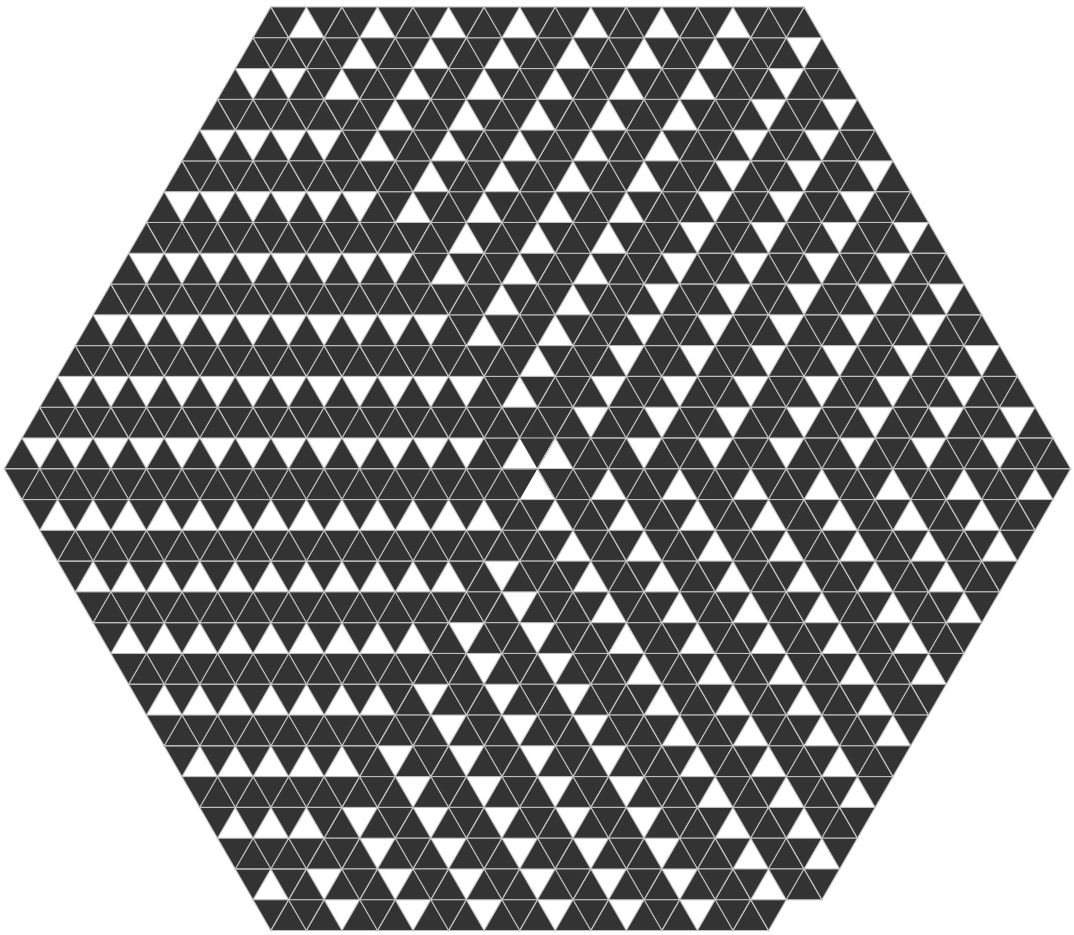


Figure 4: T_{315}

2 Preliminary Results

Define the perimeter $p(A)$ of a polyiamond to be the number of edges that are on the topological boundary of A. For all integers $n \geq 1$ we denote by $p_{\min}(n)$, the minimum perimeter possible for polyiamonds with n tiles. In 1976 [5], Harary and Harborth proved that

$$p_{min}(n) = 2\left\lceil \frac{1}{2}(n + \sqrt{6n}) \right\rceil - n. \tag{5}$$

Furthermore, they showed that the polyiamond in $\mathcal{T}_n$ in which tiles are placed in a minimal hexagonal spiral attains $p_{min}(n)$ for every n (See Figure 3 in [5] and Figure 13 in [8]).

Although p_{min} is not a monotone function, the increments are always $+1$ or -1. This fact appears as a comment along with Figure 13 in [8], and we include it as part of the following lemma.

Lemma 2.1. *For any natural number n, $p_{min}(n+1) = p_{min}(n) \pm 1$. And if $k \geq 1$, then $p_{min}(n) - 1 \leq p_{min}(n+k) \leq p_{min}(n) + k$.*

Proof. As noted in [8], in the hexagonal spiral the $(n+1)$-th tile shares one edge with the n-th tile, and if it is pointing outward from the center then it shares a second edge with a tile that was placed in the previous layer of the spiral. Consider, for example, the penultimate tile on the right in the second row from the bottom of T_9 (or of any T_h) in Figure 3; this would be the last tile placed in a clockwise spiral and is outward facing, whereas the one to its right is inward facing.

So if the $(n+1)$-th tile is pointing outward, it closes two edges of the perimeter while adding one, and

$$p_{min}(n+1) = p_{min}(n) - 1.$$

Conversely, if the $(n+1)$-th tile is pointing inward, it only shares the one connecting edge with the n-th tile. So it closes one while adding two, and

$$p_{min}(n+1) = p_{min}(n) + 1.$$

Thus $p_{min}(n+k) - p_{min}(n) \leq k$, since each additional tile can add at most one edge to the perimeter.

Along a given side of this hexagonal shape, the tiles alternate between outward and inward, except at the corners where two inward facing tiles are placed one after the other (see Figure 13 in [8]). Hence along a given side, p_{min} alternates between values $i-1$ and i, and along the next side it alternates between i and $i+1$. In particular, once the threshold i is reached by $p_{min}(n)$, it can never drop below $i-1$ as n increases. ■

The number of edges which are on the boundary of two tiles of a polyiamond A will be denoted by $b(A)$. These are the edges contained in the

interior of A. Observe that all the edges of the individual tiles of A necessarily either belong to the perimeter or to the interior of A. This means that

$$3n = p(A) + 2b(A). \tag{6}$$

Let $b_{min}(n)$ be the minimum number of interior edges that a polyiamond with n tiles can have. We associate a dual graph to a polyiamond by considering each polygon as a vertex and by connecting any two of these vertices if they share an edge, so that $b(A)$ is precisely the number of edges in this graph.

Since polyiamonds have connected interiors by definition, the associated dual graph of a polyiamond is connected and has a spanning tree. Then the dual graph of a polyiamond with n tiles has n vertices, and its spanning tree has $n-1$ edges. Hence $b_{\min}(n) \geq n-1$. Observing that the polyiamond C in which n tiles are placed in a row has exactly $b(C) = n-1$, we conclude that $b_{min}(n) = n-1$.

To distinguish which edges on the perimeter bound holes in a polyiamond A, we define the hole perimeter $p_h(A)$ to be the number of edges bounding holes, and the remaining outer perimeter $p_{out}(A)$ to be the number of edges which do not bound holes, so that

$$p(A) = p_{out}(A) + p_h(A). \tag{7}$$

If a polyiamond A is simply connected, then $p(A) = p_{out}(A)$. Using these parameters, we formulate a necessary condition for the existence of a polyiamond with n tiles and h holes.

Lemma 2.2. *If there exists a polyiamond with n tiles and h holes in which each hole has an area of 1, then*

$$h \leq \frac{3n - 2b_{min}(n) - p_{min}(n+h)}{3}. \tag{8}$$

Proof. Let A be a polyiamond with n tiles and h holes, each with an area of 1. Then by (6) and (7),

$$p_h(A) = 3n - 2b(A) - p_{out}(A). \tag{9}$$

Since $b(A) \geq b_{min}(n)$, and by filling in each hole we observe that $p_{out}(A) \geq p_{min}(n+h)$, we have that

$$p_h(A) \leq 3n - 2b_{min}(n) - p_{min}(n+h). \tag{10}$$

Holes in polyiamonds have at least three edges, so finally,

$$h \leq \frac{p_h(A)}{3} \leq \frac{3n - 2b_{min}(n) - p_{min}(n+h)}{3}.$$

■

For convenience, we define notation for this upper bound function,

$$\begin{aligned} M(n,h) &= \frac{3n - 2b_{min}(n) - p_{min}(n+h)}{3} \\ &= \frac{n + 2 - p_{min}(n+h)}{3}. \end{aligned} \tag{11}$$

Lemma 2.2 asserts that if a polyiamond exists with n tiles and h holes, each with an area of 1, then $M(n,h) \geq h$. And conversely if $M(n,h) < h$, then no such polyiamond can exist. So we examine $M(n,h)$ as a function of n, and consider the set of n for which $M(n,h) \geq h$.

Lemma 2.3. *The function $M(n,h)$ is monotonically increasing in n.*

Proof. Let $i \geq 1$ be an integer. Then applying Lemma 2.1 to $M(n+i,h)$,

$$\begin{aligned} M(n+i,h) &= \frac{n+i+2+2-p_{min}(n+h+i)}{3} \\ &\geq \frac{n+i+2-p_{min}(n+h)-i}{3} \\ &\geq \frac{n+2-p_{min}(n+h)}{3} \\ &= M(n,h). \end{aligned}$$

■

In Section 3.3, we show that the polyiamonds T_h satisfy

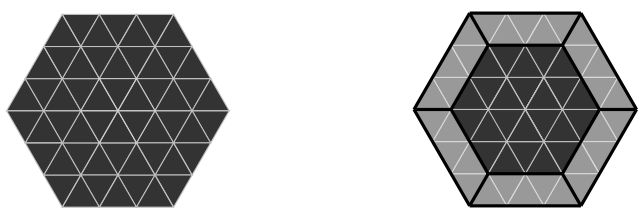
$$|T_h| = \min\{n : M(n,h) = h\}, \tag{12}$$

making $|T_h|$ optimal if all holes have an area of 1, and thus a potential candidate for $g_\triangle(h)$.

3 Proof of Theorem 1.1

3.1 Basic Shapes in the Triangular Lattice

We begin by introducing some terminology to describe arrangements of polyiamonds. We take the equilateral triangular lattice in the plane, oriented so that each triangle has a horizontal edge. We call a triangle an *up*-triangle if the horizontal edge is at the base, and a *down*-triangle if the horizontal edge is on top. Let Hex_k be the regular hexagon in this lattice with side length k, and refer to the concentric hexagonal layers as $L_k = \mathrm{Hex}_k - \mathrm{Hex}_{k-1}$. Hex_k and L_k are polyiamonds themselves, but will primarily be used as metric benchmarks in this lattice to describe the positions of the tiles of a given polyiamond. Since we only consider free polyiamonds, these are all assumed to be centered at the same origin.



Hex_3 $\qquad$ $\mathrm{Hex}_3 = \mathrm{Hex}_2 \cup L_3$

Figure 5: Central hexagons and layers, with L_3 divided into trapezoids.

For every $k \geq 1$, L_k decomposes into 6 trapezoidal row polyiamonds each containing k up-triangles and $k-1$ down-triangles, or vice versa (see Figure 5). Thus,

$$\begin{aligned} |L_k| &= 6(k+k-1) = 12k-6, \\ |\mathrm{Hex}_k| &= \sum_{i=1}^{k}(12i-6) = 12\binom{k+1}{2} - 6k = 6k^2. \end{aligned} \tag{13}$$

Recall the sequences n_k and h_k from (3), with $n_k + h_k = 6k^2 - 2$. These sequences will be, respectively, the maximum number of holes that can be created within Hex_k and the minimum number of tiles necessary to do so. The h_k terms are also necessary for defining $g_\triangle(h)$, which will be a piecewise function defined consistently for $h_k \leq h \leq h_{k+1}$. The difference terms in

these sequences are

$$\begin{aligned} h_{k+1} - h_k &= 3\binom{k+1}{2} - 3\binom{k}{2} \\ &= \frac{3}{2}\left((k+1)k - k(k-1)\right) \\ &= 3k. \end{aligned} \tag{14}$$

And

$$\begin{aligned} n_{k+1} - n_k &= \frac{9}{2}(k+1)^2 + \frac{3}{2}(k+1) - 2 - \left(\frac{9}{2}k^2 + \frac{3}{2}k - 2\right) \\ &= \frac{9}{2}k^2 + 9k + \frac{9}{2} + \frac{3}{2}k + \frac{3}{2} - \frac{9}{2}k^2 - \frac{3}{2}k \\ &= 9k + 6. \end{aligned} \tag{15}$$

That is, in the $(k+1)$-th hexagonal layer L_{k+1}, we will create $3k$ holes by adding $9k+6$ triangles. Six of these holes, occurring at the corners of the hexagon, will require four tiles, and the rest will require only three.

3.2 Constructing Crystallized Polyiamonds

Starting with the central configuration T_3, we construct the remainder of the sequence $\{T_h\}_{h\geq 4}$ by successively adding one of the three building blocks $\mathcal{A}$, $\mathcal{B}$, and $\mathcal{C}$ shown in Figure 6, with appropriate rotations, clockwise around the center.

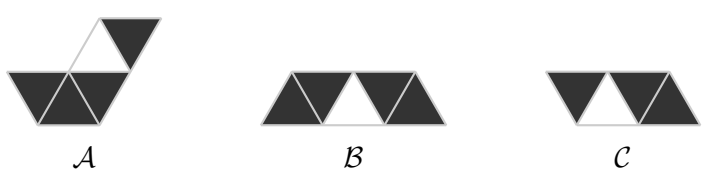


Figure 6: $\mathcal{A}$, $\mathcal{B}$, and $\mathcal{C}$ blocks used to build crystallized polyiamonds.

In the $(k+1)$-th layer, either $\mathcal{A}$ or $\mathcal{B}$ is used to turn each of the 6 corners, and $\mathcal{C}$ is used to extend along the current side of Hex_k being covered. The following rules indicate when to use each block, as depicted in Figure 7.

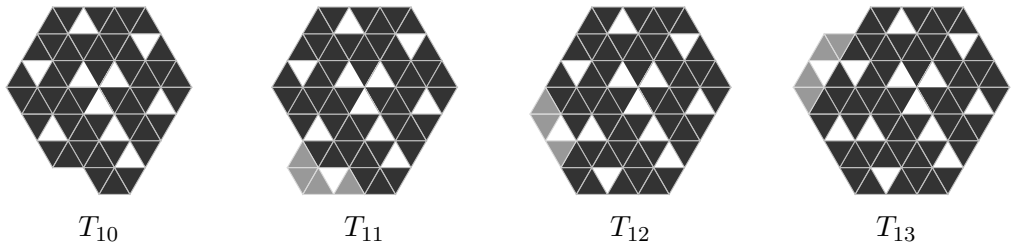


Figure 7: From T_{10} to T_{13}, first $\mathcal{A}$ is added, then $\mathcal{C}$, and then $\mathcal{B}$.

- Use $\mathcal{A}$ when only one edge of the current side of Hex_k remains uncovered. $\mathcal{A}$ is always placed with its isolated tile adjacent to the open edge of the previous block (T_{11} in Figure 7).
- Use $\mathcal{B}$ to start a new side when the previous block finishes by covering the last edge of a side. $\mathcal{B}$ is placed with one of the two tile edges on its long side on the open edge of the previous block, extending along the next side of Hex_k (T_{13} in Figure 7).
- Use $\mathcal{C}$ if at least two edges of the current side of Hex_k remain uncovered. Like $\mathcal{A}$, $\mathcal{C}$ is always placed with its isolated tile adjacent to the open edge of the previous block (T_{12} in Figure 7).

Using $\mathcal{C}$ adds three tiles and one hole, while $\mathcal{A}$ and $\mathcal{B}$ add a fourth tile. In the $(k+1)$-th layer, $\mathcal{A}$ and $\mathcal{B}$ will occur at multiples of $k/2$ when k is even, and if k is odd they alternate at intervals of $(k-1)/2$ and $(k+1)/2$. To make this precise, enumerate the holes in the $(k+1)$-th hexagonal layer with consecutive indices starting with 1, and let $w_k(l)$ denote the index of the l-th $\mathcal{A}$ or $\mathcal{B}$ in this layer.

Lemma 3.1. *For fixed $k \geq 2$ and $l \in \{1, 2, 3, 4, 5, 6\}$,*

$$w_k(l) = 1 + \left\lfloor (l-1)\frac{k}{2} \right\rfloor.$$

Proof. We prove this assuming that the first block in each layer is $\mathcal{A}$, which we use induction to establish. Starting with T_3 from Figure 1 where $k = 2$, the block needed to turn the corner is $\mathcal{A}$. Then for fixed $k \geq 3$ assume the first block in the $(k+1)$-th layer is $\mathcal{A}$, and thus $w_k(1) = 1$.

This initial $\mathcal{A}$ covers the first two of the k boundary edges on this side of Hex_k, one of which actually bounds the hole that $\mathcal{A}$ creates. Any further $\mathcal{A}$'s

used in this layer, however, will only cover one boundary edge on the next side. Meanwhile, $\mathcal{B}$ and $\mathcal{C}$ always cover two boundary edges, with $\mathcal{B}$'s first edge adjacent to the previous block, and thus in the interior of L_{k+1} instead of bordering Hex_k. These characterizations are all evident in Figure 7, where T_{10} is also constructed from T_9 (Figure 3) by adding the initial $\mathcal{A}$ for the fourth layer.

Following this first $\mathcal{A}$, there are then $\lfloor (k-2)/2 \rfloor$ many $\mathcal{C}$'s. If k is odd there is an open edge remaining and this is followed by $\mathcal{A}$, and if k is even it is followed by $\mathcal{B}$. Thus

$$w_k(2) = 1 + \left\lfloor \frac{k-2}{2} \right\rfloor + 1 = 1 + \left\lfloor \frac{k}{2} \right\rfloor.$$

On the next side, if k is odd there is only one edge covered by $\mathcal{A}$, and then $(k-1)/2$ many $\mathcal{C}$'s perfectly covering the boundary on this side, followed by $\mathcal{B}$. Then for k odd, $\lfloor k/2 \rfloor = (k-1)/2$, and

$$\begin{aligned} w_k(3) &= 1 + \left\lfloor \frac{k}{2} \right\rfloor + (k-1)/2 + 1; \\ &= 1 + (k-1)/2 + (k-1)/2 + 1; \\ &= 1 + k. \end{aligned}$$

If k is even, then $\mathcal{B}$ covers the first two edges, and $(k-2)/2$ many $\mathcal{C}$'s are placed, perfectly covering the boundary of this side, and thus followed again by $\mathcal{B}$. Then for k even, $\lfloor k/2 \rfloor = k/2$, and

$$\begin{aligned} w_k(3) &= 1 + \left\lfloor \frac{k}{2} \right\rfloor + (k-2)/2 + 1; \\ &= 1 + \frac{k}{2} + \frac{k}{2}; \\ &= 1 + k. \end{aligned}$$

The next side starts with $\mathcal{B}$ in both cases, which covers two edges of the new side. This is precisely what the initial $\mathcal{A}$ on the first side did, and so the above increments repeat for the next two sides, proving the formula for $w_k(4)$ and $w_k(5)$. Then the pattern of the first side repeats for the fifth side, proving $w_k(6)$, and it only remains to show that the next layer must also begin with $\mathcal{A}$.

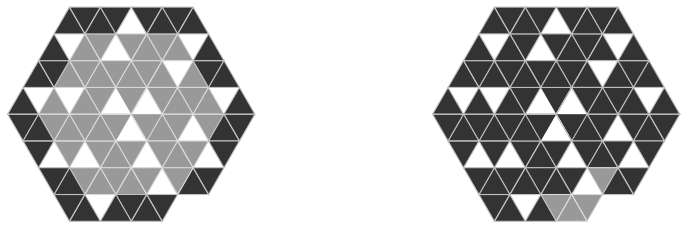


Figure 8: T_{18} shown with its L_4 layer perfectly covering Hex_3 at left, and the uncovered edge of the initial $\mathcal{A}$ in L_4 on the sixth side at right.

Observe that the sixth side in L_{k+1} follows the same pattern as the second side, and so this side of Hex_k is perfectly covered by the succession of $\mathcal{C}$'s. However, as this last side transitions into L_{k+2}, there is also an edge from the initial $\mathcal{A}$ in L_{k+1} which needs to be covered, as depicted in Figure 8. Therefore the next block, which is the first of L_{k+2}, must be $\mathcal{A}$, and hence by induction every layer starts with $\mathcal{A}$ and the equation holds for all $k \geq 2$. ■

Lemma 3.2. $|T_{h_k}| = n_k$.

Proof. We prove this by induction. For the base case $k = 2$, we have $h_2 = 3$ and $|T_3| = 19 = n_2$ (Figure 1). For fixed $k \geq 3$, assume that $|T_{h_k}| = n_k$.

In building the $(k+1)$-th layer, we add $|L_{k+1}| = 12k + 6$ tiles and holes combined, starting with the last two tiles in L_k and ending right before the last two in L_{k+1}. In the construction, we add $x + 6$ holes and $3(x + 6) + 6$ tiles, where x is the number of $\mathcal{C}$ blocks. Hence $4(x + 6) + 6 = 12k + 6$, so this adds $x + 6 = 3k$ holes and $3(3k) + 6 = 9k + 6$ tiles. Then by equations (14) and (15), we have that

$$T_{h_k+3k} = T_{h_{k+1}},$$
$$|T_{h_{k+1}}| = n_k + 9k + 6 = n_{k+1}.$$

By induction the statement holds for all $k \geq 2$. ■

When the h-th hole is in the $(k+1)$-th layer, there are $h - h_k$ blocks in this layer, and we keep track of the extra tiles added by the $\mathcal{A}$ and $\mathcal{B}$ blocks by counting $\lceil (h - h_k)/(k/2) \rceil$. Consider that

$$w_k(l) - 1 = \left\lfloor (l-1)\frac{k}{2} \right\rfloor \leq (l-1)\frac{k}{2} \quad \Longrightarrow \quad \frac{w_k(l) - 1}{\frac{k}{2}} \leq l - 1.$$

So for $h - h_k < w_k(l)$, $\lceil (h-h_k)/(k/2) \rceil \leq l-1$. And

$$w_k(l) = 1 + \left\lfloor (l-1)\frac{k}{2} \right\rfloor \geq \frac{1}{2} + (l-1)\frac{k}{2} \quad \Longrightarrow \quad \frac{w_k(l)}{\frac{k}{2}} \geq \frac{1}{k} + l - 1.$$

Hence for $h - h_k \geq w_k(l)$, $\lceil (h-h_k)/(k/2) \rceil \geq l$. Thus

$$w_k(l) \leq h - h_k < w_k(l+1) \quad \Longrightarrow \quad \lceil (h-h_k)/(k/2) \rceil = l.$$

For such an h, three tiles are added for each of the $h - h_k$ holes, plus an extra 1 for each of the l corner blocks that have been placed.

Then for $h \geq 3$, $|T_h|$ is given by the piecewise function defined on the intervals $3\binom{k}{2} \leq h \leq 3\binom{k+1}{2}$ by

$$\begin{aligned} |T_h| &= n_k + 3\,(h - h_k) + \left\lceil \frac{h - h_k}{\frac{k}{2}} \right\rceil \\ &= n_k + 3\left(h - \left(\frac{3}{2}k^2 - \frac{3}{2}k\right)\right) + \left\lceil \frac{2\left(h - \frac{3}{2}\left(k(k-1)\right)\right)}{k} \right\rceil \\ &= 3h + 6k - 2 + \left\lceil \frac{2h}{k} \right\rceil - 3(k-1) \\ &= 3h + 3k + 1 + \left\lceil \frac{2h}{k} \right\rceil. \end{aligned} \tag{16}$$

3.3 Crystallization of T_h

To complete the proofs of Theorems 1.1, 1.2, and 1.4, we first prove equation (12), showing equality in Lemma 3.3, and minimality in Lemma 3.4. We then rule out the possibility that $g_\triangle(h) < |T_h|$ using Lemmas 2.1 and 2.2, and finally examine the possible structure for crystallized polyiamonds.

Observe that in Figure 1, we have polyiamonds T_1, T_2, and T_3 with 1, 2, and 3 holes, and 9, 14, and 19 tiles, respectively. The following are easily verified by plugging into equation (11), and they assert that equation (12) holds for $h \in \{1, 2, 3\}$:

$$M(9,1) = 1, \qquad M(14,2) = 2, \qquad M(19,3) = 3,$$

$$M(8,1) = \frac{1}{3}, \qquad M(13,2) = \frac{4}{3}, \qquad M(18,3) = \frac{7}{3}.$$

Lemma 3.3. $M\left(|T_h|, h\right) = h$.

Proof. The first three cases are shown above. We then prove this by induction, from the base case $M(19, 3) = 3$. Assume that $M\left(|T_{h-1}|, h-1\right) = h-1$ for some $h > 3$. Then

$$\begin{aligned} 3(h-1) &= |T_{h-1}| + 2 - p_{min}\left(|T_{h-1}| + h - 1\right) \\ \implies \qquad 3h &= \left(|T_{h-1}| + 3\right) + 2 - p_{min}\left(|T_{h-1}| + h - 1\right) \end{aligned} \tag{17}$$

Recall that for $h > 3$, T_h is built from T_{h-1} by adding one of the $\mathcal{A}$, $\mathcal{B}$, and $\mathcal{C}$ blocks. Via the proof of Lemma 2.1, and evident from Figure 13 in [8], the increments in p_{min} as the individual tiles for these respective blocks are added one at a time are

$$\begin{aligned} &\mathcal{A} : +1, -1, +1, +1, -1; \\ &\mathcal{B} : +1, +1, -1, +1, -1; \\ &\mathcal{C} : +1, -1, +1, -1. \end{aligned}$$

Hence when an $\mathcal{A}$ or $\mathcal{B}$ block is used, then $|T_h| = |T_{h-1}| + 4$ and

$$p_{min}\left(|T_h| + h\right) = p_{min}\left(|T_{h-1}| + h - 1\right) + 1.$$

Thus equation (17) gives

$$\begin{aligned} 3h &= \left(|T_{h-1}| + 3\right) + 2 - \left(p_{min}\left(|T_h| + h\right) - 1\right) \\ &= \left(|T_{h-1}| + 4\right) + 2 - p_{min}\left(|T_h| + h\right) \\ &= |T_h| + 2 - p_{min}\left(|T_h| + h\right). \end{aligned} \tag{18}$$

And if a $\mathcal{C}$ block is used, then $|T_h| = |T_{h-1}| + 3$ and

$$p_{min}\left(|T_h| + h\right) = p_{min}\left(|T_{h-1}| + h - 1\right).$$

So equation (17) gives

$$\begin{aligned} 3h &= \left(|T_{h-1}| + 3\right) + 2 - p_{min}\left(|T_h| + h\right) \\ &= |T_h| + 2 - p_{min}\left(|T_h| + h\right). \end{aligned} \tag{19}$$

Equations (18) and (19) are equivalent to $M\left(|T_h|, h\right) = h$, and hence this holds for all h by induction. ■

Lemma 3.4. $M\left(|T_h|-1,h\right)<h$.

Proof. Each of the $\mathcal{A}$, $\mathcal{B}$, and $\mathcal{C}$ blocks end with an outward facing tile, so by the proof of Lemma 2.1 we have that

$$p_{min}\left(|T_h|+h\right)=p_{min}(|T_h|-1+h)-1.$$

Applying this equality and Lemma 3.3, we have that

$$\begin{aligned}
M\left(|T_h|-1,h\right) &= \frac{(|T_h|-1)+2-p_{min}\left(|T_h|-1+h\right)}{3} \\
&= \frac{|T_h|+1-p_{min}\left(|T_h|+h\right)-1}{3} \\
&= \frac{|T_h|-p_{min}\left(|T_h|+h\right)}{3} \\
&< M\left(|T_h|,h\right)=h.
\end{aligned}$$

■

In fact, it is clear from the proof that $M\left(|T_h|-1,h\right)=h-2/3$, as observed directly for $h\in\{1,2,3\}$. Together, Lemma 3.3 and 3.4 imply that $|T_h|=\min\{n:M(n,h)=h\}$, and we are ready to prove Theorem 1.1.

Proof of Theorem 1.1. Fix a positive integer h. Equation (16) matches that in Theorem 1.1, so by the existence of T_h it suffices to show that if A is a polyiamond with $h(A)=h$, then $|A|\geq|T_h|$.

Suppose that A is a polyiamond with $h(A)=h$, and that each hole in A has an area of 1. Then $M(n,h)\geq h$ by Lemma 2.2, and by equation (12) $|T_h|$ is the minimum such n, so $|A|\geq|T_h|$.

Now suppose that A is a polyiamond with $h(A)=h$ and at least one hole of area at least 2, and by way of contradiction suppose further that $|A|=n<|T_h|$. Then the total area of the holes of A can be written as $h+i$, and the hole perimeter $p_h(A)=3h+j$ for some integers $i,j\geq 1$. Similar to the reasoning of the proof in Lemmas 2.2, we have that

$$p_h(A)=3h+j\leq n+2-p_{min}(n+h+i).$$

And by Lemma 3.3, for T_h we have

$$3h=|T_h|+2-p_{min}(|T_h|+h).$$

Combining these two relations for h we get

$$\begin{aligned} & |T_h| + 2 - p_{min}(|T_h| + h) + j \leq n + 2 - p_{min}(n + h + i) \\ \iff \quad & |T_h| - n + j \leq p_{min}(|T_h| + h) - p_{min}(n + h + i). \end{aligned}$$

Observe that by adding tiles to fill in the extra space in the holes of A, we can create a polyiamond A' in which each hole has an area of 1, and $|A'| = n+i$. But by equation (12), T_h has the fewest tiles amongst polyiamonds with h holes each having an an area of 1. Hence $|T_h| \leq n+i$, and $|T_h|+h \leq n+h+i$. Therefore, by Lemma 2.1,

$$\begin{aligned} & p_{min}(|T_h| + h) - p_{min}(n + h + i) \leq 1, \\ \implies \quad & |T_h| - n + j \leq 1. \end{aligned}$$

But by assumption both $|T_h| - n$ and j are at least 1, so $|T_h| - n + j \geq 2$ and this is a contradiction. Therefore $|A| \geq |T_h|$, and $|T_h| = g_\triangle(h)$. ■

Corollary 3.4.1. *For every $h \geq 1$, the polyiamond T_h is crystallized.*

The inequalities in this proof also help to establish that a crystallized polyiamond does not have any holes with area greater than 1, which we then use to prove the other two conditions in Theorem 1.4.

Proof of Theorem 1.4. In the above proof of Theorem 1.1, if $n = |T_h|$ and all else remains the same, then the last inequality simplifies to $j \leq 1$. This only happens if $j = i = 1$ and there is exactly one hole of area 2, which has four edges in its perimeter instead of three. It also requires that

$$p_{min}(|T_h| + h) - p_{min}(|T_h| + h + 1) = 1 \tag{20}$$

By the proof of Lemma 2.1, this only happens if the $(|T_h| + h + 1)$-th tile is outward facing. However, each of the $\mathcal{A}$, $\mathcal{B}$, and $\mathcal{C}$ blocks end with an outward facing tile, which are always followed by an inward facing tile, and thus (20) is never satisfied. Hence if a polyiamond is crystallized, then every hole has an area of 1.

Note that as a corollary of Theorem 1.1 and Lemma 3.3, if A is crystallized, then $M(|A|, h(A)) = h(A)$. And so if $A \in \mathcal{T}_n$,

$$p_h(A) = 3h(A) = 3n - 2b_{min}(n) - p_{min}(n + h(A)).$$

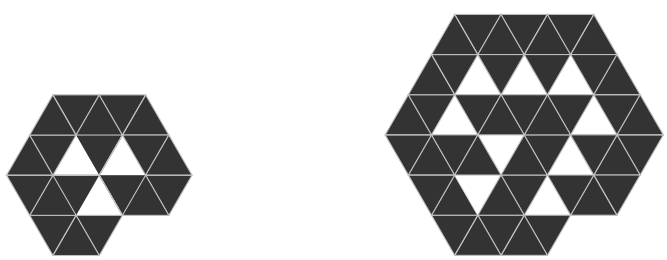
Suppose that A is crystallized. If the dual graph of A is not a tree, then it contains at least n edges and $b(A) \geq n > n-1 = b_{min}(n)$. Similarly, if A does not achieve the minimum outer perimeter, then because every hole has an area of 1, $p_{out}(A) > p_{min}(n + h(A))$. In either case, by equation (9)

$$p_h(A) < 3n - 2b_{min}(n) - p_{min}\left(n + h(A)\right),$$

which is a contradiction, and hence A cannot be crystallized. ■

4 Crystallized Spirals

In this section we construct a sequence of crystallized spiral polyiamonds $\{Spir_k\}_{k\geq 2}$. The smallest case is $Spir_2 = T_3$, and for all other h and k the polyiamonds $Spir_k$ and T_h are distinct. $Spir_k$ is constructed as follows inductively from $Spir_2$ and $Spir_3$ in Figure 9.



$Spir_2 = T_3$ $\qquad$ $Spir_3 \ncong T_9$

Figure 9: Central crystallized spirals for even and odd k.

The k-th element in the sequence is built from $Spir_{k-2}$ by first adding the last two tiles in the bottow right corner of L_{k-2}. Then add all down-triangles in L_{k-1}, and all of L_k except the two tiles in its bottom right corner. Next, to ensure the interior is connected, take the penultimate up-triangle in the bottom row of L_{k-2} and shift it down and to the right, to the penultimate up-space in the bottom row of L_{k-1} (see Figure 10).

This creates $Spir_k$, a spiral of holes with central $Spir_2$ and $Spir_3$ configurations, depending on whether k is even or odd, respectively. Furthermore, for all $k \geq 2$, $Spir_k$ is crystallized.

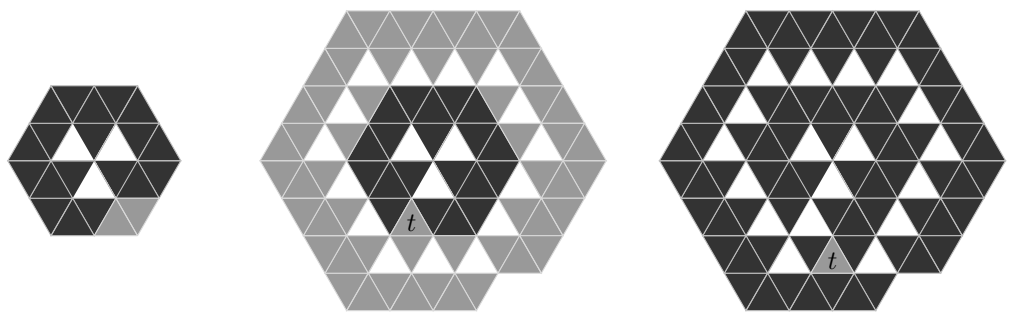


Figure 10: From left to right: Fill in bottom corner of L_2 in $Spir_2$; add all down-triangles in L_3 and all but the bottom right two tiles in L_4; finally move the tile t from L_2 to L_3 to get $Spir_4$.

Lemma 4.1. *For $k \geq 2$, $Spir_k$ is crystallized with $h(Spir_k) = h_k$ and $|Spir_k| = n_k$.*

Proof. In this construction, all holes outside of the central configuration are up-triangles. Triangles in a layer are evenly split between up and down, so L_{k-1} contains $6k - 9$ up-triangles by equation (13). In building $Spir_k$ from $Spir_{k-2}$, every up-triangle in L_{k-1} becomes a hole except for the one which is filled by the tile from L_{k-2} to connect the spiral, which of course creates one up-triangle hole in L_{k-2} that was not there before. So the number of holes added in this process is precisely the number of up-triangles in L_{k-1}. Thus $h(Spir_k) = h(Spir_{k-2}) + 6k - 9$.

From Figure 9 it is clear that $h(Spir_2) = 3 = h_2$ and $h(Spir_3) = 9 = h_3$. Fix $k \geq 4$ and assume that the formula holds for all values less than k. Then $h(Spir_{k-2}) = h_{k-2}$, and

$$\begin{aligned} h(Spir_k) &= h_{k-2} + 6k - 9 \\ &= \left(\frac{3}{2}(k-2)^2 - \frac{3}{2}(k-2)\right) + 6k - 9 \\ &= \frac{3}{2}k^2 - 6k + 6 - \frac{3}{2}k + 3 + 6k - 9 \\ &= \frac{3}{2}k^2 - \frac{3}{2}k. \end{aligned}$$

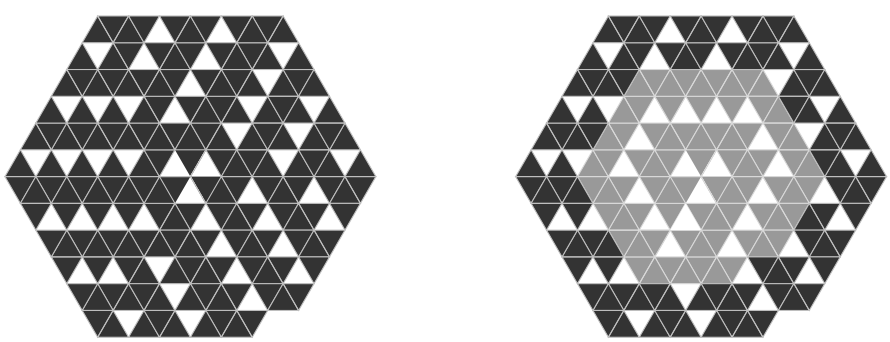
By induction, $h(Spir_k) = h_k$ for all integers $k \geq 2$. And these polyiamonds fill out all of Hex_k except for the bottom right two triangles, so

$$|Spir_k| = 6k^2 - 2 - h_k = n_k.$$

Thus $g_\triangle\left(h\left(Spir_k\right)\right) = |Spir_k|$, and $Spir_k$ is cyrstallized. ■

Proof of Theorem 1.3. For $3 \leq h \leq 8$, examples of distinct crystallized polyiamonds are given in Figure 2. For $h \geq 9$, observe that T_{h_k} has the same outer perimeter as $Spir_k$, and its central configuration is always $Spir_2$. Hence for $h \geq h_K$, the central configuration of T_h can be swapped out with $Spir_k$ for any $3 \leq k \leq K$ to create distinct crystallized polyiamonds (see Figure 11). ■

Definition 4.1. For $h \geq h_k$, let $T_h * Spir_k$ denote the polyiamond in which T_h exchanges its central configuration for $Spir_k$. If $h = h_k$, then $T_{h_k} * Spir_k = Spir_k$.



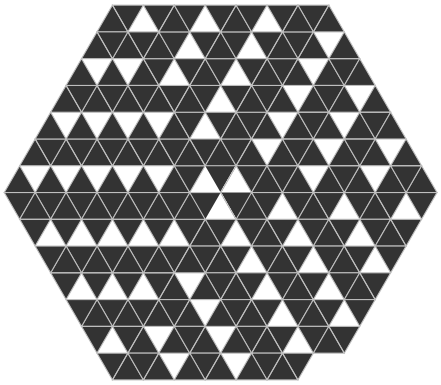
T_{45} $T_{45} * Spir_4$

Figure 11: The center of T_{45} is exchanged with $Spir_4$.

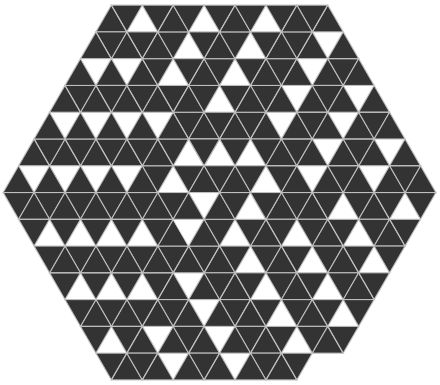
Corollary 4.1.1. *For $k \geq 3$ and $h \geq h_k$, there are at least $k - 1$ distinct crystallized polyiamonds with h holes, given by the set*

$$\{T_h\} \cup \{T_h * Spir_l : 3 \leq l \leq k\}.$$

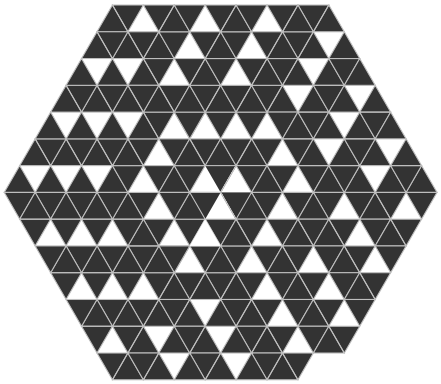
This set is depicted in full for $h = h_7 = 63$ in Figure 12, and in Figure 13 we take T_{315} from Figure 4 and form $T_{315} * Spir_8$.



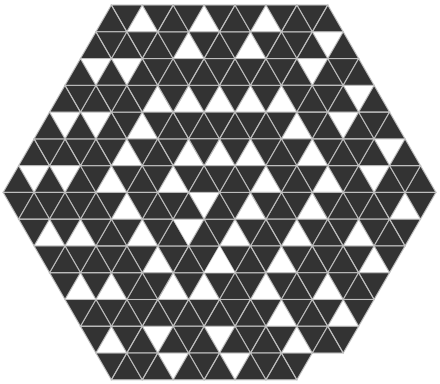
T_{63}



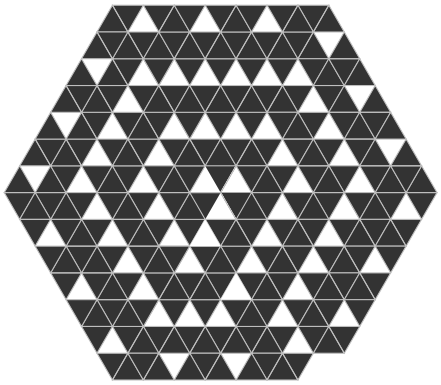
$T_{63} * Spir_3$



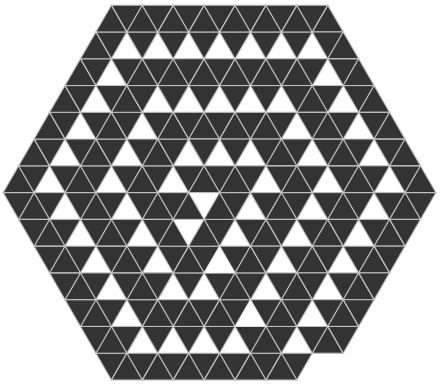
$T_{63} * Spir_4$



$T_{63} * Spir_5$



$T_{63} * Spir_6$



$Spir_7$

Figure 12: Six distinct crystallized polyiamonds with 63 holes.

5 Concluding Remarks

We have completely determined the sequences $g_\triangle(h)$ and $f_\triangle(n)$ and given several necessary conditions for the crystallization of a polyiamond. Perhaps the only remaining open question regarding the crystallization of polyiamonds is that of enumeration for each $h \geq 3$, and the structural characterization in Theorem 1.4 narrows the scope of this question considerably.

It is also clear that changing the basic shape of the tiles can have a drastic impact on the nature of this analysis. As discussed in Section 1.1, similar methods to those employed here produce only a partial solution for polyominoes. In [7], it is also proved by constructive geometric arguments that $g_\square(h)$ is the right inverse of $f_\square(n)$, that $f_\square(n+1) - f_\square(n) \leq 1$, and that $g_\square(h) = n$ if and only if $f_\square(n) = h$ and $f_\square(n-1) = h-1$. These are all results which we get with polyiamonds as a direct consequence of proving that the T_h are crystallized. It is not clear, however, that this will be true for any choice of polygon, especially those which do not tile the plane.

A further generalization of 2-dimensional polyforms is constructed in [6], in which the tiles are again regular n-gons for some choice of n, but each tile is required to share edges with at least k other tiles (in the definition of polyforms, $k = 1$). These structures are called (n, k)-scatters. The restriction of having a finite number of tiles is removed, and an (n, k)-scatter is said to be crystallized if there exists an infinite shape which necessarily appears in any (n, k)-scatter. In particular, $(5, 3)$-scatters crystallize, and we conjecture that one of the tessellations with finite symmetry groups (see Figure 12 in [6]) gives a way of constructing a sequence of polyforms with regular pentagons as tiles that will crystallize in the sense of Definition 1.3.

Acknowledgments.

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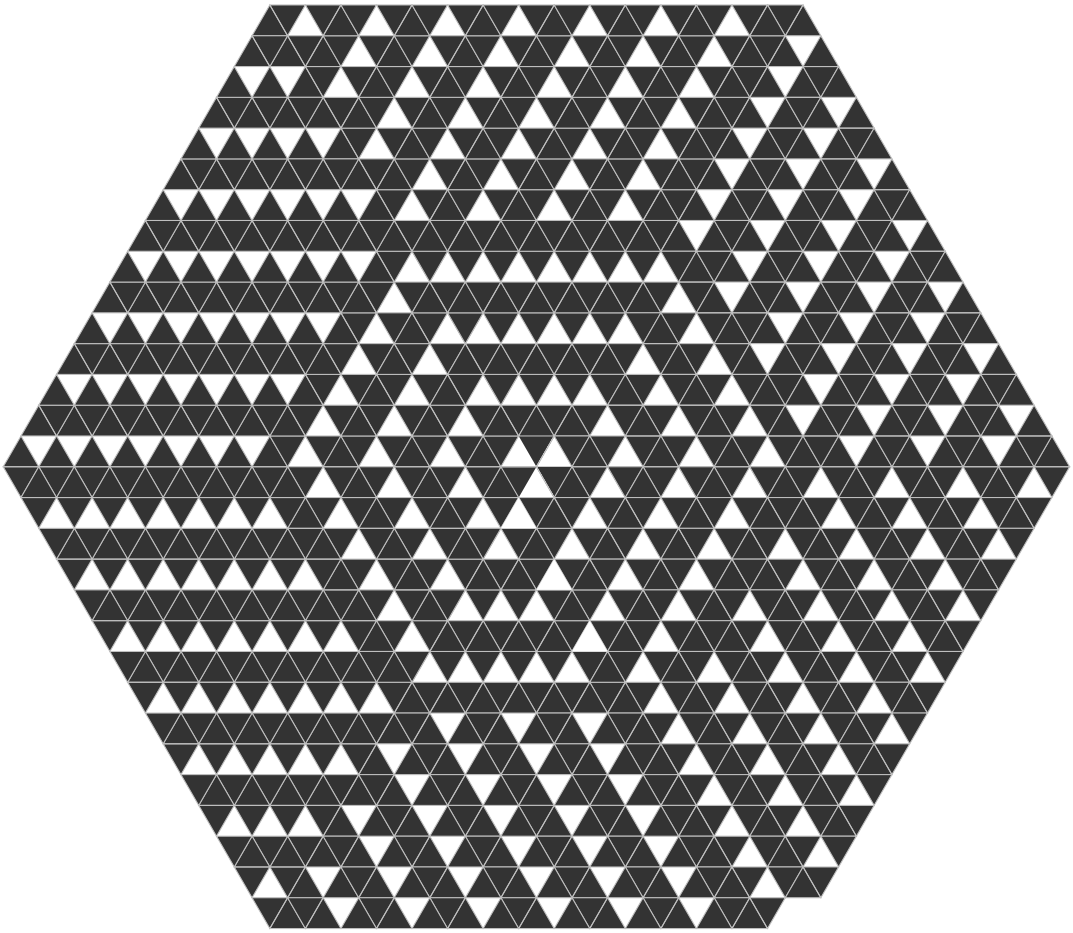

Figure 13: $T_{315} * Spir_8$

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